HydroSense

Scientific Computing Approach for Smart Irrigation Management

Introduction

HydroSense addresses efficient irrigation in Kenya by combining data processing, numerical analysis, and simulation to improve water use and crop outcomes.

Explain challenges: *unpredictable rainfall*, water scarcity, inefficient irrigation, and lack of **real-time monitoring** necessitate computational support for data-driven irrigation.

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Topic 1

Project Overview

HydroSense integrates environmental datasets, numerical methods, and visualization to support irrigation decisions.

The system produces cleaned datasets, ET estimates, and optimized irrigation guidelines using reproducible Python workflows and scientific validation.

Motivation & Challenges

Kenyan agriculture faces variable rainfall, limited water resources, and sensor inconsistencies.

Challenges include missing data, outliers, file management, and maintaining numerical accuracy for reliable irrigation advice.

Objectives & Scope

Load and preprocess agricultural data, detect sensor faults, apply numerical methods, and compare performance between implementations.

Deliverables: cleaned datasets, ET calculations, validated numerical solvers, and performance benchmarks for ICS 2207 requirements.

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Topic 2

Datasets (Weather, Soil)

Weather: temperature, rainfall, humidity, wind, solar index.

Soil: moisture, tank level, pump flow, sensor status.

Crop zones: irrigation needs and crop parameters for decision support.

Data Quality Issues

Common issues: missing timestamps, inconsistent units, abnormal readings (impossible tank levels, out-of-range temperatures), and pump faults.

Data cleaning emphasizes transparency, documenting decisions and preserving scientific meaning while restoring dataset usability.

Cleaning Techniques

Implement *missing value handling* via interpolation and median replacement to preserve trends.

Use IQR-based outlier detection and rule-based *sensor fault detection* for impossible readings (e.g., negative tank level).

Document cleaning steps and maintain reproducible logs for auditability and transparency.

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Topic 3

Computational Methods

Apply root-finding (Bisection, Newton-Raphson, Secant), numerical differentiation, and integration for irrigation modelling.

Use Runge-Kutta and Euler schemes for soil moisture simulation and Monte Carlo for rainfall uncertainty assessment.

Python Implementation

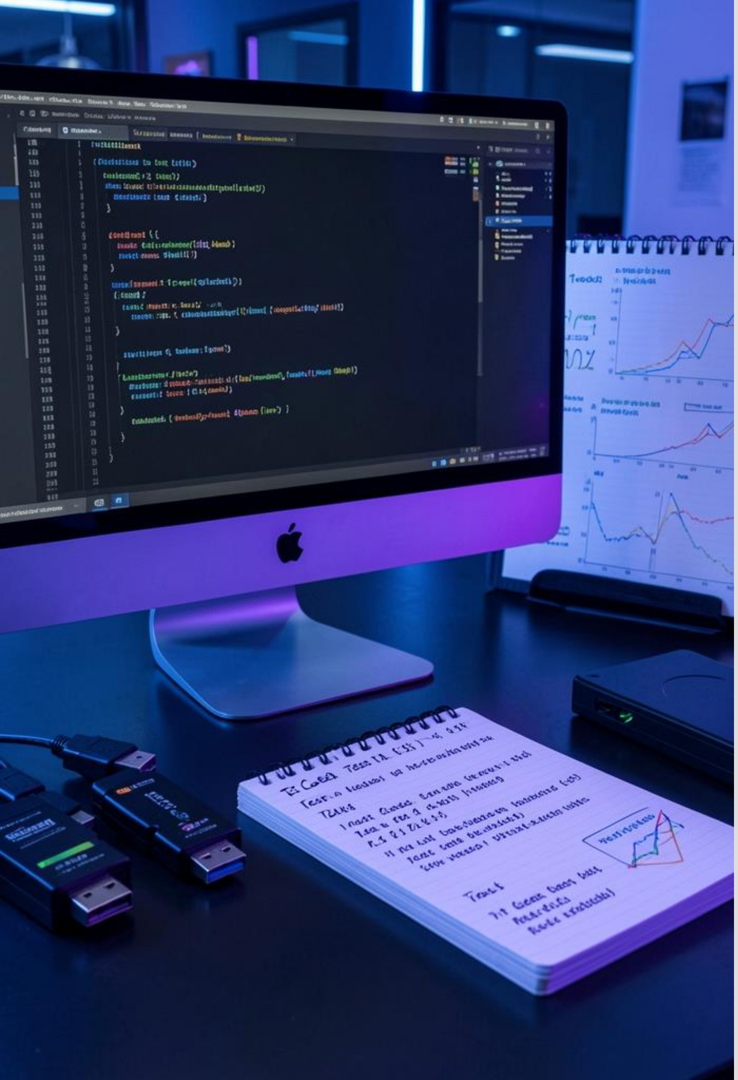
Modular codebase using Python with **Pandas**, **NumPy**, Matplotlib, and PyTest.

Organize: src/data_cleaning.py, src/numerical_methods.py, tests/test_root_finding.py for maintainability and reproducibility.

Performance Comparison

Compare Python loop vs NumPy vectorized ET calculation: measure execution time and verify numerical equivalence.

Vectorization yields significant speedups through optimized array operations and reduces computation overhead.



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Topic 4

Testing & Validation

Use *PyTest* to verify root-finding accuracy, convergence behavior, and error handling.

Unit tests validate cleaning procedures, numerical solvers, and reproducibility of simulation outputs.

Results & Visuals

Deliverables include cleaned irrigation datasets, completed ET maps, and performance charts.

Present **bar chart** for execution time, time-series plots for soil moisture, and decision-support visual summaries.

Future Improvements

Integrate machine learning for irrigation prediction and *real-time IoT* sensor ingestion.

Develop a mobile dashboard, implement weather-forecast coupling, and automate model retraining for continuous improvement.

Conclusions

HydroSense demonstrates a reproducible scientific computing pipeline that cleans environmental data, applies numerical methods, and optimizes irrigation decisions.

The project shows improved computational efficiency via vectorization, validated numerical solvers, and practical deliverables for smart irrigation management.